

	COMPASS Synoptic Field Sampling Protocol	
Date Created: 4 Apr 2022 Date Updated: March 2023	Creator Name(s): A. Stearns Editor Name(s): A. Stearns	Version: 1
Tree Plot Installation & Annual Census		

Objective:

To establish a tree vegetation plot that will allow COMPASS to monitor a representative sample of trees over time.

Contents:

- I. Experimental Design
- II. Personal Protective Equipment
- III. Materials
- IV. Procedure
 - A. Plot Installation
 - B. Census
- V. Site Specific Notes

I. Experimental Design:

Each COMPASS Synoptic Site will have two (2) tree plots installed, one (1) in the Transition Zone and one (1) in the Upland Zone. Every tree located within the tree plot will be monitored annually for growth and condition status. Ideal tree plots will include as many sap flux trees as possible.

In 2024 Sweet Hall Marsh added an additional zone, the Swamp. Officially, the originally (2023) designated Upland became Swamp, and the originally (2023) designated Upland Control became Upland where a new tree plot was installed and inventoried.

II. Personal Protective Equipment:

Close-toed shoes and long pants are required at all times when working at COMPASS sites. Garden or work gloves are recommended for hand safety. Other PPE, such as eye and ear protection, is readily available if desired.

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III. Materials:

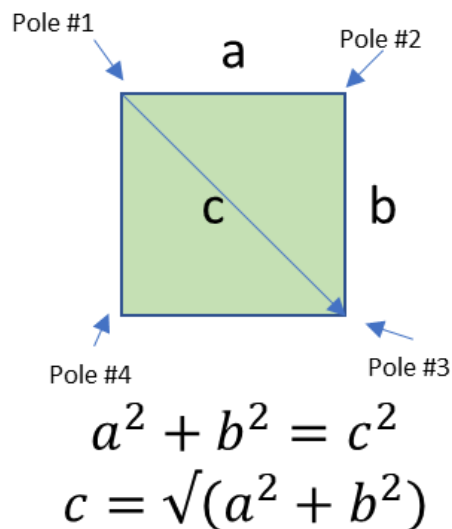
- (2) 30M+ metric survey tapes
- 5M D-tape
- 1" PVC corner poles - 4/plot or 8/site
- Mallet
- Crayons (DBH marker)
- Tags
- 2" aluminum nails (make sure they fit the tag hole diameter)
- Hammer
- Data collection sheet
- Clipboard
- Pencil

IV. Procedure:

A. Establishing the Plot

1. Designate the general area of the plot.
2. Set corner pole #1
3. Pull tape in the direction you want the distance you want the width
4. Set corner pole #2
5. From corner pole #2, pull tape to the distance desired for the length, approximating a right angle, in the proposed direction of pole #3
6. Use the Pythagorean Theorem to find the hypotenuse

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7. Pull tape from pole #1 toward the proposed position of pole #3
8. Place pole #3 where the length of the side intercepts the length of the hypotenuse
9. Pull tapes from pole #'s 1 and 3 to find the placement of the last corner, pole #4

B. Census

1. Mark the Height of Measure (HOM) on the south side of the tree using a wax crayon.
 - a. The HOM should be taken at DBH, 1.3M from the ground surface.
 - b. If the tree is on a slope, the HOM should be taken 1.3M from the ground surface of the *uphill* side of the tree.
2. Using a metric diameter tape, measure and record the diameter at the designated HOM
3. Tree tags should be placed on the south side of the tree (opposite side of the sap flow sensors) and approximately 10 cm above or below the HOM (so as not to interfere with the HOM due to potential tree swelling around the nail). Be sure NOT to hammer the nail in too deep in order to leave room for tree growth. This is why a 2" or longer nail is ideal.
4. Status Codes:
 - LI = Living
 - DS = Dead standing
 - DC = Dead down

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DN = Dead/not found

V. Site Specific Notes: